

Research

Risk management in the production phase of oil and gas projects: evaluating risk responses and their impact on safety and damage reduction

Mohammad Khalilzadeh¹ · Sayyid Ali Banihashemi² · Ali Heidari³ · Seyed Mohammad Hossein Orazani⁴ · Peyman Taebi⁵

Received: 27 January 2025 / Accepted: 23 April 2025

Published online: 12 May 2025

© The Author(s) 2025 [OPEN](#)

Abstract

Energy projects play a critical role in modern human societies, significantly influencing economic growth and societal welfare. However, they are often criticized for imposing substantial environmental burdens. This dual impact underscores the necessity of implementing effective risk management strategies to identify, evaluate, and mitigate potential risks. Such strategies not only aim to minimize environmental consequences but also optimize project costs and timelines by leveraging advanced assessment techniques, innovative methodologies, and cutting-edge technologies. This study focuses on the exploitation phase of a selected energy project, employing a multi-faceted approach. Risks are systematically identified through a comprehensive literature review, organizational insights, and expert judgment. The identified risks are then filtered using the Fuzzy Delphi method to ensure relevance and accuracy. High-priority risks are determined through a probability-impact matrix, followed by the application of the Fuzzy Best–Worst Method (FBWM) to weigh and prioritize criteria for risk response planning. Finally, the Grey Combined Compromise Solution (CoCoSo –G) method is utilized to rank and select the most effective risk response strategies under uncertainty. The findings provide valuable insights and practical guidelines that can be leveraged in similar energy and construction projects. By facilitating informed decision-making, this research aims to assist managers and decision-makers in formulating robust risk response plans tailored to uncertain and dynamic project conditions.

Keywords Risk management · Decision-making · Fuzzy Delphi · Fuzzy Best–Worst Method (FBWM) · CoCoSo–G Method · Uncertainty

1 Introduction

The oil and gas industry stands as one of the most influential global sectors, significantly shaping national power and reputation while serving as a cornerstone for job creation and economic stability [1]. Beyond its direct contributions, this industry intersects with numerous other sectors, including construction, and plays a pivotal role in both

* Ali Heidari, ali.heidari1990@ut.ac.ir; Mohammad Khalilzadeh, khalilzadeh@srb.iau.ac.ir; Sayyid Ali Banihashemi, banihashemi@pnu.ac.ir; Seyed Mohammad Hossein Orazani, smh.orazani@gmail.com; Peyman Taebi, peymantaebi@gmail.com | ¹Department of Industrial Engineering, Science and Research Branch, Islamic Azad University, Tehran, Iran. ²Department of Industrial Engineering, Payame Noor University, Tehran 193954697, Iran. ³School of Industrial Engineering, Faculty of Engineering, University of Tehran, Tehran, Iran. ⁴Department of Industrial Engineering, Qazvin Branch, Islamic Azad University, Qazvin, Iran. ⁵School of Management and Economics, Science and Research Branch, Islamic Azad University, Tehran 1477893855, Iran.



the development of industrial ecosystems and the welfare of human societies [2, 3]. Annually, billions of dollars are allocated worldwide to oil and gas projects, emphasizing their economic and strategic importance. However, these projects are not without challenges. Political, regional, and environmental factors consistently influence oil and gas endeavors, leading to significant risks that threaten natural ecosystems, human health, and project outcomes. Such risks, if unaddressed, can derail project objectives and compromise their success [4]. Consequently, robust risk management strategies are essential to identify, evaluate, and mitigate these risks. Employing innovative tools and methodologies, such as Building Information Modeling (BIM) and the Internet of Things (IoT), further enhances the likelihood of project completion within defined budgetary and temporal constraints. Given the unique characteristics of oil and gas projects, particularly their complexity and scale, recognizing and addressing risks is imperative. This necessitates a comprehensive approach to understanding the most effective risk management strategies, assessing their applicability, and measuring their effectiveness in real-world scenarios. While risk management in this sector often prioritizes health, safety, and hygiene [5–7], broader areas such as equipment condition and operational risks receive comparatively less attention. In this study, we focus on a critical oil and gas project located in southern Iran, characterized by its EPCC (Engineering, Procurement, Construction, and Commissioning) nature and daily oil production capacity. This project exemplifies the frequent neglect of risk management during the operational phase. Addressing this gap, the research aims to:

Identify and evaluate significant project risks.

Select optimal risk response plans based on expert judgment and established criteria. Assess the impact of implementing these plans under uncertain conditions.

Unlike prior studies that primarily focused on identifying risks or selecting response plans, this research integrates these phases with the evaluation of response plan effectiveness. Moreover, it introduces advanced multicriteria decision-making (MCDM) techniques, including the Best–Worst Method (BWM) for criteria weighting and the Combined Compromise Solution (CoCoSo–G) approach for ranking and selecting risk response strategies. By leveraging grey and fuzzy triangular numbers, this study addresses the inherent uncertainty of risk evaluation in oil and gas projects. The study's novelty lies in its holistic approach to managing project risks and its application of cutting-edge methodologies. Previous research often failed to simultaneously address fuzzy risks, evaluate multiple response plans, and utilize CoCoSo–G and related logic-based methods to prioritize actions and measure their post-implementation impacts.

The structure of this paper is as follows: Sect. 2 reviews the relevant literature and contextualizes the current research within the broader risk management landscape. Section 3 details the research methodology, criteria, parameters, and model framework. Section 4 presents the results, providing a step-by-step analysis of the case study and its application of the proposed methods. Finally, Sect. 5 concludes the study, summarizing its contributions and implications.

2 Literature review

In previous literature, risk is known as an unidentified circumstance regarding an unknown future that complicates further prediction of the results of a project, with uncertainty being the main core [1]. Active avoidance and resolution of project risks are vital for project continuity, as proper reactions and achieving predefined goals demand a specific capability approach. All projects have associated risks due to their unique properties, which can significantly influence profitability, costs, timing, final quality, and beneficiary satisfaction [2, 3]. Consequently, risk management must persist throughout all phases of a project's lifecycle to ensure suitable control and mitigate negative effects on project outcomes.

The foundation of risk management lies in employing a systematic approach to addressing risks and uncertainties, aiming to enhance the quality of project preparation and evaluation [4]. Risk management plays a pivotal role in achieving organizational goals across commercial aspects by improving efficiency, effectiveness, organizational competence, and sustainability [5], ensuring durability, profitability, and project completion [6]. Furthermore, it enables organizations to respond effectively to emerging opportunities and threats [7]. Risk analysis provides a comprehensive overview of system performance under uncertain conditions [8], prevents conflicts among project parties and beneficiaries, minimizes time



and cost losses due to unforeseen conditions, protects shareholders from value loss, and steers projects from failure to success. Additionally, integrating risk management with quality management boosts organizational profitability by identifying potential threats in product manufacturing and design, thereby enhancing overall organizational performance [9]. Risk management encompasses not only identifying and evaluating risks but also responding to them, which is critical during the project's operational phase [10, 11]. Risk response serves as a direct mechanism for modifying risks, reducing their impact during action phases, and optimizing project performance [12]. It involves decision-making that requires a deep understanding of risks and implementing necessary steps to minimize their occurrence and effects, ultimately reducing uncertainty and enhancing project success [13]. The primary objective of risk response is to formulate, select, and implement strategies [14], addressing risks that threaten the project's trajectory and existence by eliminating or transferring them [15, 16]. Success in executing project risk management heavily relies on this step and should not be underestimated, as the risk management plan is only effective when the risk response plan is enacted across all project components [17]. This process demands selecting the appropriate methods to overcome uncertainties [18], ensuring acceptable project performance [19]. Population growth, economic development, the global importance of nuclear energy [20], and environmental degradation exacerbate resource risks. Moreover, the rising demand for meeting safety standards amidst increasing risks of serious incidents [21] underscores the urgency for effective risk management and timely responses worldwide [22]. Safeguarding employees' health against workplace risk factors and maximizing their physical, mental, and social well-being require coordinated efforts among governments, employers, employees, and other stakeholders. Governments must ensure the enforcement of laws and regulations, while employers and employees are responsible for creating and maintaining safe working conditions [23]. Risk evaluation encompasses multiple areas such as human health, asset loss, and environmental pollution, which are often not aligned. Comprehensive assessment and control of overall risks remain complex tasks, particularly for non-experts in fields like environmental risk management [24]. Nguyen and Machion systematically reviewed 64 studies on green construction risks, risk evaluation models, and risks in green refinement projects, emphasizing the need for robust risk management strategies in this domain [25]. Similarly, other research explored critical risks in green construction projects, analyzing participant relations and risk evaluation using regression analysis, providing valuable insights for stakeholders to adopt effective risk management approaches [26]. With the increasing emphasis on clean energy production, such as wind energy, many countries are expanding their wind farm capacities. Zhao et al. studied design risks in wind electricity projects, analyzing investment impacts and proposing models for addressing construction risks through proactive measures [27]. Failure to execute risk management techniques exacerbates uncontrolled cost and time risks, jeopardizing profitability and stakeholder trust while increasing the likelihood of project failure [28]. Consistent and precise risk management throughout the project lifecycle ensures survival, profitability, and the ability to respond effectively to emerging challenges and opportunities [6, 7]. Correct investment decision-making, a cornerstone of business sustainability, relies on identifying risks and evaluating financial plans. Senova et al. [4] introduced a Ball Crystal simulation approach using the Monte Carlo method for classifying investment risks and predicting time deviations under uncertainties. Elravvad et al. used an Analytical Hierarchical Process (AHP) to identify financial risks in micro and medium businesses, enhancing decision-making reliability [5]. Traditional risk response planning involves recognizing, evaluating, and recommending strategies based on past practices. However, modern approaches emphasize scale and criteria in response planning to optimize resource allocation and ensure effective risk mitigation [29]. The Combined Compromise Solution (CoCoSo) method, a flexible MCDM technique, is particularly effective in handling uncertain and imperfect data [30]. Panpatil et al. [31] applied Pythagoras fuzzy hierarchical models and CoCoSo for evaluating green supply chain management strategies, focusing on pollution reduction and environmental sustainability. Kursk et al. utilized PF-LBWA and PF-CoCoSo methods to select smart network strategies for the logistics industry, enhancing environmental safety [32]. In addressing African transportation challenges, Burayma et al. combined IRN SWARA and IRN CoCoSo methods for railway system optimization [33]. Applications of CoCoSo and related techniques span various fields, such as flood forecasting [34], cloud service selection [35], IoT-enabled waste management in smart cities [36], and share selection in financial decision-making [30]. Banihashemi and Khalilzadeh used fuzzy BWM and CoCoSo to evaluate rural water supply projects [37]. Tavana et al. combined BWM and CoCoSo for environmental package selection under unpredictable conditions [38]. Adar et al. applied AHP and CoCoSo to waste management, aiding policymakers in decision-making [39]. BIM (Building Information Modelling) and IoT are increasingly applied in construction risk management for their data reliability, 3D visualization, cost control, and project lifecycle benefits [40–43]. Recent studies highlight their role in managing risks in oil and gas infrastructure projects, enabling organizations to enhance safety and efficiency [44–50]. A brief review of relevant research works is summarized in Table 1.

Table 1 Summary of the related studies

Author	Research aim		Research methodology				Industry under study				
	Risk management fields		Type of number		MCDM methods		Oil and Gas industry	Other industries			
	Year	Risk identification and assessment	Risk response	Investigation of the effectiveness of risk response strategies	Crisp	Fuzzy			Grey	Hybrid compromise solution	TOPSIS, VIKOR and the other MCDM methods
Goh & Abdul-Rahman [13]	2013										✓
Nanthuru et al. [3]	2018	✓	✓								✓
Tanojo et al. [17]	2018	✓				✓					✓
Guimaraes et al. [21]	2018	✓	✓								✓
Yazdani et al. [54]	2019							✓			✓
Mokhtari & Aghagoli [7]	2020	✓	✓								✓
Renault et al. [2]	2020	✓	✓								✓
Mirsharafeddin [10]	2020	✓	✓			✓			✓		
Chu & Wang [14]	2020	✓	✓				✓				
Yücelgazi & Yitmen [15]	2020	✓	✓						✓		✓
Qin et al. [24]	2020	✓	✓				✓				✓
Chou & Chiu [1]	2021	✓	✓						✓		
Gudipati & Cha [8]	2021	✓	✓			✓					✓
Zhang & Guan [11]	2021	✓	✓			✓					✓
Li et al. [20]	2021		✓								✓
Qian & Liang [22]	2021	✓	✓			✓			✓		✓
AK et al. [23]	2021	✓	✓			✓			✓		✓
Toh & Zhang [28]	2021	✓	✓								✓
Brown et al. [29]	2021	✓	✓								✓
Tabatabaee et al. [46]	2021	✓	✓				✓		✓		✓
Yan et al. [6]	2022	✓	✓				✓		✓		✓
Narang et al. [30]	2022	✓						✓			✓
Mandal & Ali Khan [35]	2022					✓		✓			✓
Mishra et al. [36]	2022						✓	✓			✓
Tavana et al. [38]	2022						✓	✓			✓
Adar et al. [39]	2022							✓		✓	✓
Alrawad et al. [5]	2023	✓						✓			✓
Alqurashi et al. [9]	2023	✓									✓
Nguyen & Macchion [25]	2023	✓									✓
Nguyen et al. [26]	2023										✓
Panpatil et al. [31]	2023					✓		✓			✓
Korucuk et al. [32]	2023							✓		✓	✓

)

Tabl1 continued

Author	Research aim		Research methodology			Industry under study	
	Risk management fields		Type of number	MCDM methods		Oil and Gas industry	Other industries
	Year	Risk identification and assessment		Crisp Fuzzy Grey	Hybrid solution compromise the other methods		
Bouraima et al [33]	2023 ✓				✓		✓
Tabarestani et al [34]	2023				✓		
Banihashemi Khalilzadeh [35]	2023 ✓		✓		✓		✓
Zhu et al [40]	2023 ✓			✓	✓		✓
This study	2024 ✓	✓		✓	✓	✓	

- Understanding and managing risks in projects, especially in complex industries such as construction, renewable energy, and supply chain management, has become a fundamental requirement for success. Recent studies, focusing on various innovations, demonstrate that employing advanced approaches such as multi-criteria decision-making (MCDM) models, hybrid methods like CoCoSo and AHP, and fuzzy concepts can significantly enhance accuracy and predictability in risk assessment and management.
- Moreover, leveraging Building Information Modeling (BIM) and Internet of Things (IoT) technologies has revolutionized project risk management. These technologies not only enable better information management and risk reduction throughout the project lifecycle but also improve capabilities such as 3D modeling, cost control, environmental impact assessment, and design change management.
- Research also indicates that linking classical risk management methods to data-driven innovations (e.g., Monte Carlo simulation and probabilistic modeling) and applying modern decision-making techniques significantly reduce complexities associated with risk assessment across diverse fields, including wind farm construction, waste management, and transportation infrastructure. This assists companies in identifying not only existing threats but also emerging opportunities more effectively.
- Overall, recent research demonstrates that integrating hybrid frameworks, fuzzy approaches, advanced decisionmaking methods, and cutting-edge technologies such as BIM and IoT represents a transformative trend in risk management. These innovations not only enhance organizational efficiency and effectiveness in addressing risks but also provide a foundation for better decision-making under uncertainty and achieving greater project sustainability.

As summarized in Table 1, a significant research gap exists in the domain of project risk management. While previous studies have primarily focused on the development of risk response strategies, very few have extended their scope to evaluate these strategies and assess their effectiveness post-implementation. This highlights a critical shortfall in understanding the practical impact of such strategies in real-world applications. Furthermore, the CoCoSo method, a novel multi-criteria decision-making (MCDM) approach, has seen limited application in project risk management, despite its potential to offer more robust and precise decision-making under uncertainty [51–53]. Existing literature lacks comprehensive studies that address the following:

Identification and evaluation of risks specific to the exploitation phase of oil and gas projects.

Formulation and prioritization of risk response strategies using advanced methodologies.

Application of innovative decision-making techniques, such as CoCoSo and grey fuzzy logic, for selecting the most appropriate risk responses.

Post-implementation evaluation of the effectiveness of risk response strategies in achieving project objectives.

This study fills this gap by proposing a comprehensive framework that integrates systematic risk identification, advanced evaluation methods, and innovative decision-making approaches. It is one of the first to implement CoCoSo in conjunction with grey fuzzy logic for risk management in the oil and gas industry, offering a novel perspective on improving the accuracy and effectiveness of risk response planning.

3 Materials and methods

3.1 Managing uncertainty in fuzzy inputs at different stages of analysis

In risk assessment and response planning for oil and gas projects, uncertainty is a critical factor that can impact decisionmaking processes. The integration of fuzzy logic into risk analysis methods, such as the Fuzzy Delphi Method (FDM) and Fuzzy Best–Worst Method (FBWM), provides a structured approach to handling uncertainty by incorporating expert opinions and linguistic variables. This section outlines how uncertainty from fuzzy inputs is transferred and managed across different stages of the analysis.

3.1.1 Uncertainty in risk identification

The first stage of the analysis involves identifying potential risks associated with the oil and gas production phase.

Given the complexity and variability of risks in such environments, expert judgments are solicited through the Fuzzy Delphi Method. The uncertainty in expert opinions is captured using triangular fuzzy numbers, which represent linguistic assessments such as "highly significant," "moderately significant," and "low significance." These fuzzy representations enable the aggregation of diverse perspectives while maintaining flexibility in defining risk importance [25, 26, 37].

3.1.2 Uncertainty in risk prioritization

Once risks are identified, they are prioritized based on their probability and impact using a fuzzy probability-impact matrix. The inherent uncertainty in estimating these parameters is addressed by using fuzzy logic to convert qualitative judgments into quantitative values. The fuzzy membership functions ensure that variations in expert opinions do not lead to abrupt changes in prioritization but instead provide a smooth transition between different risk levels [1, 8].

3.1.3 Uncertainty in criteria weighting

The next stage involves determining the relative importance of different risk response criteria using the Fuzzy Best-Worst Method (FBWM). Decision-makers select the best and worst criteria, and pairwise comparisons are conducted using fuzzy linguistic terms. To manage uncertainty in these comparisons, a fuzzy pairwise comparison matrix is constructed, ensuring that variations in expert judgment are incorporated into the weighting process. The optimal fuzzy weights are derived by solving a constrained optimization problem, minimizing inconsistencies in decision-making [7, 11].

3.1.4 Uncertainty in risk response selection

Risk response strategies are evaluated using the Combined Compromise Solution with Grey Systems (CoCoSo-G) method, which integrates fuzzy logic and grey numbers to address data imperfections. The uncertainty in selecting optimal responses is mitigated by normalizing fuzzy-weighted criteria and aggregating them into a decision matrix. This approach ensures that imprecise and incomplete information does not lead to suboptimal risk response planning [31, 38, 39].

3.1.5 Uncertainty in effectiveness evaluation

Finally, the effectiveness of selected risk responses is assessed under uncertain conditions. The responses are simulated under various operational scenarios, and their performance is evaluated using fuzzy-set-based sensitivity analysis. By considering multiple possible outcomes, decision-makers can better understand the resilience of each risk response plan and adapt strategies to mitigate emerging risks effectively [27, 32, 33].

Managing uncertainty in risk analysis is crucial for ensuring the reliability of decision-making processes in oil and gas projects. The integration of fuzzy logic into risk identification, prioritization, criteria weighting, response selection, and effectiveness evaluation provides a comprehensive framework for addressing uncertainty. This structured approach enhances the robustness of risk management strategies and improves the adaptability of decision-making under dynamic project conditions. By leveraging advanced methodologies such as FBWM and CoCoSo-G, this study contributes to the growing body of research on uncertainty management in industrial projects, aligning with previous works by Senova et al. [4] and Nguyen et al. [26].

3.2 The Delphi method

Ishikawa et al. introduced the fuzzy Delphi method [55] for combining the fuzzy set theory and Delphi method for facing uncertainties. It includes these three following stages:

Stage 1: Identifying indices using literature review and theoretical fundamentals of the research.

Stage 2: Collecting the expert opinions (decision-maker experts): After recognizing indices, a questionnaire is set for studying identified indices and delivered to each expert for screening and study of indices. The linguistic variables and triangular fuzzy numbers shown in Table 2 are used to state the significance of the indices.

Table 2 Linguistic terms and fuzzy Delphi numbers

Linguistic variables	Triangular fuzzy numbers
Very low	(0,0,0.25)
low	(0,0.25,0.5)
Average	(0.25,0.5,0.75)
High	(0.5,0.75,1)
Very high	(0.75,1,1)

Stage 3: Confirmation and screening of the indices: In this stage, indices are compared with the S threshold value, and the indices lower than the threshold value are not confirmed and removed from the main research indices. There is no scientific and legitimate method for defining threshold value and defined through intellectual inference of the decision-maker and directly affects the number of screened parameters. In this research, experts selected 0.5 as the threshold value. Triangular fuzzy values of experts' opinions are calculated for screening indices and the fuzzy average is computed for measuring n respondent opinions. The fuzzy value τ for each index is computed based on Eqs. (1–4):

$$\tau_{ij} = (a_{ij}, b_{ij}, c_{ij}), i = 1, 2, \dots, n, j = 1, 2, \dots, m \quad (1)$$

$$a_j = \frac{\sum a_{ij}}{n} \quad (2)$$

$$b_j = \frac{\sum b_{ij}}{n} \quad (3)$$

$$c_j = \frac{\sum c_{ij}}{n} \quad (4)$$

In the abovementioned relations, the i index denotes expert and the j denotes the decision-maker index. Also, the defuzzified value of the average fuzzy values is obtained from Eq. (5) [56].

$$risp = \frac{a + b + c}{3} \quad (5)$$

3.3 The fuzzy best-worst method (FBWM)

The Best-Worst Method (BWM) is one of the multi-criteria decision-making techniques for calculating criteria weight and choices suggested by Rezaei [56]. Based on ambiguities in decision data due to imperfect data and uncertainties in the qualitative judgment of decision-makers, fuzzy concept interference in MCDM can provide a more reliable decision-making result. Hence, this research uses the BWM technique, a result of combining fuzzy set theory and BWM, to provide a more reliable weight than BWM since five main linguistic terms for conducting reference comparisons between criteria and choices. This aids decision-makers in running reference comparisons easily and accurately. Therefore, the fuzzy BWM technique provides a more consistent comparison compared with certain states from this method [57]. The pairwise comparison matrix in a fuzzy state is shown as follows:

$$\tilde{A} = \begin{bmatrix} c_1 & \tilde{a}_{11} & \tilde{a}_{12} & \dots & \tilde{a}_{1n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_n & \tilde{a}_{n1} & \tilde{a}_{n2} & \dots & \tilde{a}_{nn} \end{bmatrix} \quad (6)$$

In the above matrix, $\tilde{a}_{ij}$ is a triangular fuzzy number that indicates the fuzzy relative significance of the i th criterion as compared with the j th criterion. When $\tilde{a}_{ij} = (1, 1, 1)$, it indicates $i = j$.

In pairwise comparison matrix in a fuzzy state $\tilde{A}$, there are $2n-3$ pairwise comparisons that $n-2$ is the result of comparing the best criterion as compared with other criteria, and the other $n-2$ is the result of comparing the remaining criteria as compared with the worst criteria and $+1$ is the result of comparing the best criterion with worst criterion. Definition 1: $\tilde{a}_{ij}$ pairwise comparison is defined as fuzzy reference comparison if i is the best element or j is the worst element.

Stage 1: Generating decision criteria set: In this stage, criteria required for reaching a decision are obtained based on experts' opinions and literature review, which are shown as $\{c_1, c_2, \dots, c_n\}$.

Stage 2: Defining the best and worst criterion: In this stage, decision-makers define the best and worst criterion based on decision criteria set individually. The best criterion is denoted as c_w and the worst criterion is denoted as c_b . Comparison is not conducted in this stage.

Stage 3: Performing fuzzy reference comparison for selecting criterion: Based on definition 2, a fuzzy reference comparison includes two sections: In the first section, the pairwise comparison is $\tilde{a}_{ij}$, in which i denotes the best criterion and c_i is shown in c_b . The second section is a state, in which, the pairwise comparison is $\tilde{a}_{ij}$ and j is the worst criterion the worst criterion c_j is shown in c_w and other criteria are compared with this criterion.

The fuzzy vector of the best criterion as compared with others is obtained by using Eq. (7):

$$\tilde{A}_B = (\tilde{a}_{B1}, \tilde{a}_{B2}, \dots, \tilde{a}_{Bn}) \quad (7)$$

In fuzzy vector of the best criterion as compared with others $\tilde{A}_B$, $\tilde{a}_{Bj}$ indicates the fuzzy preference of the best criterion c_B as compared with other criteria j , $j=1, 2, \dots, n$, evidently the relation $\tilde{a}_{BB} = (1, 1, 1)$ holds.

Stage 4: Conducting the reference fuzzy comparisons for the worst criterion: Like the previous stage, the preference of the other criteria over the worst criterion is defined using linguistic variable values in the table. The fuzzy vector of the other criteria as compared to the worst criterion is defined by using Eq. (8):

$$\tilde{A}_W = (\tilde{a}_{1w}, \tilde{a}_{2w}, \dots, \tilde{a}_{nw}) \quad (8)$$

$\tilde{a}_{jw}$ is the preference of criterion j over the worst criterion c_w , $j=1, 2, \dots, n$, evidently the relation $\tilde{a}_{ww} = (1, 1, 1)$ holds. **Stage**

5: Defining optimal fuzzy weights: Eq. (9) must hold for each pair of $w_{\tilde{w}^B}$ and $\overline{w_{\tilde{w}^B}}$ must hold for the optimal fuzzy

weights of $(\tilde{W}_1^*, \tilde{W}_2^*, \dots, \tilde{W}_n^*)$:

$$\frac{\tilde{W}_j}{\tilde{W}_w} = \tilde{a}_{jw} \text{ and } \frac{\tilde{W}_j}{\tilde{W}_B} = \tilde{a}_{Bj} \quad (9)$$

The maximum absolute differences of $|\tilde{w}_{\tilde{w}^B} - \tilde{a}_{Bw}|$ and $|\tilde{w}_{\tilde{w}^B} - \tilde{a}_{jw}|$ must be minimized for all j s to meet the above condi-

tion for all j s. The limited optimization problem is obtained as

below by considering that values are not negative and fuzzy weights summation conditions: $\min \max \{ |\tilde{w}_{\tilde{w}^B} - \tilde{a}_{Bj}|, |\tilde{w}_{\tilde{w}^B} - \tilde{a}_{jw}| \}$

$$\text{s.t. } \sum_{j=1}^n \tilde{w}_j = 1, \tilde{w}_j \geq 0$$

}

$$\begin{pmatrix} | \\ | \\ w \\ n \end{pmatrix}$$

$$\begin{aligned} & R \tilde{W} = 1 \\ & \sum_{j=1}^n S. t. t_{jw}^l \leq m_{wj} \leq u_{jw} \\ & l_{jw} \geq 0 \end{aligned} \quad (10)$$

For $\tilde{W} \mid m, u, \tilde{W} \mid m, u, \tilde{W} \mid m, u, \tilde{a} \mid m, lu$, and $\tilde{a} \mid m, u$, the above equation can be transformed into the below non-linear optimization problem: $\min \tilde{\xi}^*$

$$\begin{aligned} & \tilde{W} \tilde{W}_{wjB} = \tilde{W} \tilde{W}_{wjB} \\ & \tilde{a}_{jwBj} \leq \xi_{wj} \\ & \sum_{j=1}^n \tilde{a}_{wj} \leq \xi \\ & S. t. \mid R \tilde{W} = 1 \end{aligned} \quad (11)$$

where $\tilde{\xi} \mid m, u$. By considering $l \leq m \leq u$, it is assumed that $\tilde{W} = (k, k, k)$, $k \mid$. Such that Eq. (11) is transformed into Eq. (12) based on the above assumption. $\min \tilde{\xi}^*$

$$\begin{aligned} & \left(\frac{l_{jw}^w m_{jw}^w u_{jw}^w}{l_{jw}^w m_{jw}^w u_{jw}^w} \right) \\ & \sum_{j=1}^n R \\ & \mid l_{jw} \leq m_{jw} \leq u_{jw} \\ & S. t. \tilde{W} = 1 \end{aligned} \quad (12)$$

$$\{ \mid l_j = m \mid u_j^w \}$$

By solving the above problem, criteria optimal weights in the fuzzy state of $(\tilde{W}_1^*, \tilde{W}_2^*, \dots, \tilde{W})$ can be achieved.

3.3.1 Consistency rate of the fuzzy BWM method

Comparison between criteria is consistent when the relation $\tilde{a}_{Bj} * \tilde{a}_{jw} = \tilde{a}_{Bw}$ holds for all js. In this relation, terms indicate the fuzzy importance of the best criterion as compared with criterion j, the fuzzy importance of criterion j as compared with the worst criterion, and the fuzzy importance of the best criterion as compared with the worst criterion.

The consistency ratio is calculated using the consistency index Table 3 and Eq. (13) (consistency rate). This consistency rate is in the range [0,1] and as ξ^* grows, the consistency ratio is higher and comparisons have lower reliability. Hence, if this ratio is closer to zero, comparisons have higher consistency and stability and as it approaches one, comparisons have lower consistency and stability [51].

$$\text{consistency ratio} = \frac{\xi^*}{\xi^*} \quad (13) \text{ consistency index}$$

3.4 Triangular fuzzy number (TFN)

The participation rule of the set to classic theory is whether or not each object belongs to a certain set and as a result, the range of the set and the members are evident and can be defined [58]. In contrast, in fuzzy set theory, a membership degree is chosen between 0 and 1 for each object and range of set and the members cannot be defined precisely.

Table 3 Consistency index of the BWM method

Absolutely important (AI)	Very important (VI)	Fairly important (FI)	Weakly important (WI)	Equally important (EI)	Linguistic variables
(3.5, 4, 4.5)	(2.5, 3, 3.5)	(1.5, 2, 2.5)	(0.67, 1, 1.5)	(1,1,1)	a_{Bw}
8.04	6.69	5.29	3.8	3	Consistency Index (CI)

Numbers in the abovementioned set that involve a range of values instead of referring to a specific value are called fuzzy numbers. A fuzzy that has a triangular membership function state is called a triangular fuzzy number [59].

Definition 2:

Fuzzy number $\tilde{a}$ that belongs to the real number set R is called a triangular fuzzy number such that the membership function $\mu_{\tilde{a}}(x): R \rightarrow [0,1]$ is equal to:

$$f(x) = \begin{cases} 0, & x < 1 \\ 1, & x \geq 1 \end{cases}$$

$$\tilde{a}(x)=\left\{\begin{array}{ll} \frac{x-l}{m-l} & l \leq x < m \\ \frac{u-x}{u-m} & m \leq x \leq u \\ 0, & \text{otherwise} \end{array}\right.$$

(14)

Triangular fuzzy numbers are comprised of three real numbers indicated as $\tilde{a}=(l,m,u)$. The upper bound indicated as u and lower bound indicated as l are maximum and minimum values of fuzzy number $\tilde{a}$ can take. The value m is the most probable value of a fuzzy number. Triangular fuzzy number is defined as $\tilde{a}=(l,m,u)$ [60, 61].

3.5 Defuzzification

The results of fuzzy inference from transforming fuzzy numbers into crisp numbers are simplified and comprehensible and are called defuzzification. Graded mean integration representation is used to calculate the interval between indices and comparison of criteria [54].

$$\tilde{a}_i = \frac{(l_i + 4m_i + u_i)}{6}$$

(15) 6

4 Linguistic variables

Pairwise fuzzy comparisons are performed for each criterion using fuzzy sets and nonlinguistic variables and the membership function value is shown in Table 4 [62, 63].

4.1 The CoCoSo-G method

The extended CoCoSo grey method (CoCoSo-G) was developed for solving decision-making issues in unknown environment facing uncertainty problems with unknown imperfect data [54]. The stages of this method are as follows:

- Stage 1: Criteria and choices for the decision problem of interest are defined first.
- Stage 2: X decision matrix is generated:

Table 4 Linguistic variables and their membership function value

Linguistic variables	Membership Function
Equally important (EI)	(1, 1, 1)
Weakly important (WI)	(0.76, 1, 1.5)
Fairly important (FI)	(1.5, 2, 2.5)
Very important (VI)	(2.5, 3, 3.5)
Absolutely important (AI)	(3.5, 4, 4.5)



$$U[a, b] \quad U[a, b] \quad U[a, b] \quad X = (16)$$

$$[\quad] \quad U_{a_{n1}.b_{n1}} \quad U_{a_{n2ij}.b_{n2}} \dots U_{a_{nm}.b_{nm}}$$

In the above matrix, a_{ij} indicates the lower bound and b indicates upper bound for $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$. **Stage 3:** The normalized matrix is developed based on compromise normalization equations:

For positive criterion:

$$r = U_{c_{ij}.d_{ij}} = \frac{U[a_{ij}.b_{ij}] - \min_i U[a_{ij}.b_{ij}]}{\max_i U[a_{ij}.b_{ij}] - \min_i U[a_{ij}.b_{ij}]} \quad (17)$$

For negative criterion:

$$r = U_{c_{ij}.d_{ij}} = \frac{\max_i U[a_{ijij}.b_{ij}] - U_i[a_{ij}.b_{ijij}]}{\max_i U[a_{ij}.b_{ij}] - \min_i U[a_{ij}.b_{ij}]} \quad (18)$$

Stage 4: Weighing normalized matrix and weighing power summation of comparable choices are calculated as S_i and P_i for each choice:

$$S_i = \sum_{j=1}^n (w_j U[c_{ij}.d_{ij}])$$

$$P_i = \sum_{j=1}^n (U[c_{ij}.d_{ij}]^{w_j}) \quad (19)$$

Stage 5: The relative weight of choices is calculated using the collective strategies below. In this stage, three evaluation scoring strategies are used for generating relative weights of other choices, which are obtained from Eqs. (20–22):

$$H_{ia} = h_{1ij}, h_{2ij} = \frac{P_i}{\sum_{i=1}^m P_i + S_i} \quad (20)$$

$$L_{ia} = [l_{1ij}, l_{2ij}] = \min_i \left(\frac{S_i}{\sum_{i=1}^m S_i + P_i} + \frac{P_i}{\sum_{i=1}^m P_i + S_i} \right) \quad (21)$$

$$M_{ia} = [m_{1ij}, m_{2ij}] = \frac{\lambda S_i + (1 - \lambda)(P_i)}{\lambda \max_i S_i + (1 - \lambda)(\max_i P_i)} \quad (22)$$

For $0 \leq \lambda \leq 1$.

In Eq. (22), decision-makers choose λ (normally $\lambda = 0.5$). Nevertheless, the flexibility and stability of the suggested CoCoSo can rely on different λ values. The final rating of the choices is defined using the CoCoSo-G method based on k values (for more important and better k values):

$$K_i = (H_{ia} * L_{ia} * M_{ia})^{\frac{1}{3} + \frac{1}{3}} (H_{ia} * L_{ia} * M_{ia}) \quad (23)$$

Stage 6: Different k values are calculated in this stage and an acceptable list of rated choices is obtained. Scoring vector of choices, which are grey values s and t can be transformed into clear values if s and t are larger than zero. Therefore, the maximum value of K matrix elements is defined. Then, score of each choice is obtained accordingly.

The score corresponding to Eq. (24) is called the CoCoSo score, in which K is the clear value for the assumed grey number [60, 61, 64]:

$$= t_i - s_i) \quad \frac{K(24) t_i}{K(24) t_i}$$

5 Results

5.1 Case study

The case study is about the upper areas of oil industries in western Karun oil land with light and heavy oil reserves, which have been in the operating phase for about five years.

5.1.1 Identifying and classifying risks

Several risks relevant to the oil and gas industry were identified to estimate risks, and operation phase risks between the year 2015 and June 2022 were achieved. Due to differentiation in risk classes based on the goal of this project, 4 semi-structured interviews with the experts in this area, whose characteristics are provided in Table 5, were conducted, and they were classified into three operation, manufacturing, and Environmental Health Safety Manager (HSE) sections using the Delphi method.

The total number of 137 risks from previous studies, organizational documents of the company, and interviewing field experts were identified, among which 133 risks are negative risks and four risks are positive risks of the case study operating phase. After identifying risks, a questionnaire was provided for collecting experts' opinions on initial evaluation of risks and sent to each expert. Afterwards, the one-stage Fuzzy Delphi method was used to assess the risks based on their effects on daily oil production. The risks were evaluated using the Fuzzy Delphi method, and 23 out of 137 identified risks were confirmed by the experts in Table 6 (the risks that directly affect daily oil production, and the other risks have little importance that can be moved to the watch list).

5.1.2 Risk assessment

The experts identified, and classified risks and conducted initial evaluation of risks using Delphi fuzzy method and 6 experienced industry experts assessed and prioritized the classified risks using brainstorming method with risk matrix according to cost, time, quality, safety, and validity criteria.

Based on the above Table 7, gas leakage from gas export pipelines, oil leakage from oil export pipelines and hot gas turbines, gas compressors, and blowing systems, and electricity generators are adverse risks and have higher priority. The processing unit is a positive risk with higher priority.

5.1.3 Risk response

Reducing the harmfulness and decreasing the risks with higher scores are the primary goals of risk evaluation and management; hence, a risk response plan is developed to control and monitor risks. This section studies risk response,

Risk

A hii A hii The Che Usir Usir

res
Rishi
Gas
Dil I
Dies
Dee
Air
Rishi
Wea
கிங்
மித்
மித்

[illegible]

Table T	Row	Table R	Row
1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20
21	22	23	24
25	26	27	28
29	30	31	32
33	34	35	36
37	38	39	40
41	42	43	44
45	46	47	48
49	50	51	52
53	54	55	56
57	58	59	60
61	62	63	64
65	66	67	68
69	70	71	72
73	74	75	76
77	78	79	80
81	82	83	84
85	86	87	88
89	90	91	92
93	94	95	96
97	98	99	100
101	102	103	104
105	106	107	108
109	110	111	112
113	114	115	116
117	118	119	120
121	122	123	124
125	126	127	128
129	130	131	132
133	134	135	136
137	138	139	140
141	142	143	144
145	146	147	148
149	150	151	152
153	154	155	156
157	158	159	160
161	162	163	164
165	166	167	168
169	170	171	172
173	174	175	176
177	178	179	180
181	182	183	184
185	186	187	188
189	190	191	192
193	194	195	196
197	198	199	200
201	202	203	204
205	206	207	208
209	210	211	212
213	214	215	216
217	218	219	220
221	222	223	224
225	226	227	228
229	230	231	232
233	234	235	236
237	238	239	240
241	242	243	244
245	246	247	248
249	250	251	252
253	254	255	256
257	258	259	260
261	262	263	264
265	266	267	268
269	270	271	272
273	274	275	276
277	278	279	280
281	282	283	284
285	286	287	288
289	290	291	292
293	294	295	296
297	298	299	300
301	302	303	304
305	306	307	308
309	310	311	312
313	314	315	316
317	318	319	320
321	322	323	324
325	326	327	328
329	330	331	332
333	334	335	336
337	338	339	340
341	342	343	344
345	346	347	348
349	350	351	352
353	354	355	356
357	358	359	360
361	362	363	364
365	366	367	368
369	370	371	372
373	374	375	376
377	378	379	380
381	382	383	384
385	386	387	388
389	390	391	392
393	394	395	396
397	398	399	400
401	402	403	404
405	406	407	408
409	410	411	412
413	414	415	416
417	418	419	420
421	422	423	424
425	426	427	428
429	430	431	432
433	434	435	436
437	438	439	440
441	442	443	444
445	446	447	448
449	450	451	452
453	454	455	456
457	458	459	460
461	462	463	464
465	466	467	468
469	470	471	472
473	474	475	476
477	478	479	480
481	482	483	484
485	486	487	488
489	490	491	492
493	494	495	496
497	498	499	500
501	502	503	504
505	506	507	508
509	510	511	512
513	514	515	516
517	518	519	520
521	522	523	524
525	526	527	528
529	530	531	532
533	534	535	536
537	538	539	540
541	542	543	544
545	546	547	548
549	550	551	552
553	554	555	556
557	558	559	560
561	562	563	564
565	566	567	568
569	570	571	572
573	574	575	576
577	578	579	580
581	582	583	584
585	586	587	588
589	590	591	592
593	594	595	596
597	598	599	600
601	602	603	604
605	606	607	608
609	610	611	612
613	614	615	616
617	618	619	620
621	622	623	624
625	626	627	628
629	630	631	632
633	634	635	636
637	638	639	640
641	642	643	644
645	646	647	648
649	650	651	652
653	654	655	656
657	658	659	660
661	662	663	664
665	666	667	668
669	670	671	672
673	674	675	676
677	678	679	680
681	682	683	684
685	686	687	688
689	690	691	692
693	694	695	696
697	698	699	700
701	702	703	704
705	706	707	708
709	710	711	712
713	714	715	716
717	718	719	720
721	722	723	724
725	726	727	728
729	730	731	732
733	734	735	736
737	738	739	740
741	742	743	744
745	746	747	748
749	750	751	752
753	754	755	756
757	758	759	760
761	762	763	764
765	766	767	768
769	770	771	772
773	774	775	776
777	778	779	780
781	782	783	784
785	786	787	788
789	790	791	792
793	794	795	796
797	798	799	800
801	802	803	804
805	806	807	808
809	810	811	812
813	814	815	816
817	818	819	820
821	822	823	824
825	826	827	828
829	830	831	832
833	834	835	836
837	838	839	840
841	842	843	844
845	846	847	848
849	850	851	852
853	854	855	856
857	858	859	860
861	862	863	864
865	866	867	868
869	870	871	872
873	874	875	876
877	878	879	880
881	882	883	884
885	886	887	888
889	890	891	892
893	894	895	896
897	898	899	900
901	902	903	904
905	906	907	908
909	910	911	912
913	914	915	916
917	918	919	920
921	922	923	924
925	926	927	928
929	930	931	932
933	934	935	936
937	938	939	940
941	942	943	944
945	946	947	948
949	950	951	952
953	954	955	956
957	958	959	960
961	962	963	964
965	966	967	968
969	970	971	972
973	974	975	976
977	978	979	980
981	982	983	984
985	986	987	988
989	990	991	992
993	994	995	996
997	998	999	1000

$$\begin{aligned}
 & u_4 \quad 2.5 \, l_3 \leq k^* l_3; u_4 \quad 2.5 \, l_3 \geq k^* l_3; \\
 & \left| \begin{array}{l} m_4 - 1 \\ m_5 \end{array} \right| \leq k^* m_5; m_4 - 1 \leq k^* m_5 \\
 & \geq -k^* m_5; l_4 \, 0.67 \, u_5 \leq k^* u_5; l_4 \, 0.67 \, u_5 \geq k^* u_5; \\
 & \left| \begin{array}{l} u_4 - 1.5 \\ l_5 \end{array} \right| \leq k^* l_5; u_4 - 1.5 \leq k^* l_5 \\
 & \geq -k^* l_5;
 \end{aligned}$$

$$l_1 \quad 0.67 \, u_2 \leq k^* u_2; l_1 \quad 0.67 \, u_2 \geq k^* u_2;$$

$$\left| \begin{array}{l} m_1 - 1 \\ m_2 \end{array} \right| \leq k^* m_2; m_1 - 1 \leq k^* m_2 \geq -$$

$$k^* m_2; u_1 \, 1.5 \, l_2 \leq k^* l_2; u_1 \, 1.5 \, l_2 \geq k^* l_2;$$

$$s.t \quad \left| \begin{array}{l} 3 \\ u_2 \end{array} \right| \leq 2.5 \, k^* u_2; l_3 \, 2.5 \, k^* u_2 \geq -k^* u_2; \quad (27) \quad l$$

$$m_3 \quad 3 \, m_2 \leq k^* m_2; m_3 \quad 3 \, m_2 \geq k^* m_2;$$

$$l_5 \quad 1.5 \, u_2 \leq k^* u_2; l_5 \quad 1.5 \, u_2 \geq k^* u_2; m_5$$

$$2 \, m_2 \leq k^* m_2; m_5 \quad 2 \, m_2 \geq k^* m_2; \quad u_5$$

$$2.5 \, l_2 \leq k^* l_2; u_5 \quad 2.5 \, l_2 \geq k^* l_2;$$

$$\left| \begin{array}{l} u_3 - 3.5 \\ l_2 \end{array} \right| \leq k^* l_2; u_3 - 3.5 \leq k^* l_2 \geq -k^* l_2;$$

$$\left| \begin{array}{l} \frac{1}{6} l_3 + \frac{1}{6} 4 m_3 + \frac{1}{6} u_3 + \frac{1}{6} l_4 + \frac{1}{6} 4 m_4 + \frac{1}{6} u_4 \\ \frac{1}{6} l_3 + \frac{1}{6} 4 m_3 + \frac{1}{6} u_3 + \frac{1}{6} l_4 + \frac{1}{6} 4 m_4 + \frac{1}{6} u_4 \end{array} \right| +$$

$$u_4 +$$

$$\frac{1}{6} l_5 + \frac{1}{6} 4 m_5 + \frac{1}{6} u_5 = 1$$

Research

Discover Civil Engineering (2025) 2:91 | <https://doi.org/10.1007/s44290-025-00250-8>

$$l_1 \leq m_1 \leq u_1 \quad \Bigg| \quad \Bigg| \quad l_2$$

$$\leq m_2 \leq u_2 \quad \Bigg| \quad \Bigg| \quad l_3$$

$$\leq m_3 \leq u_3$$

$$l_4 \leq m_4 \leq u_4$$

$$\quad \Bigg| \quad l_5 \leq m_5 \leq u_5$$

$$\quad \Bigg| \quad \Bigg| \quad l_1 > 0; l_2 > 0; l_3 > 0; l_4 > 0; l_5 > 0; k \geq 0$$

Criteria are ranked (as shown in Tables 10 and 11) after entering the data in the Excel program and solving the above model using Solver:

As shown in Table 12, the consistency ratio is close to zero and is acceptable.

In the Table 13 the final average of the criteria weight is presented based on experts' opinions. The personnel protection criterion (C₄₂) is the highest priority in this table. It is based on experts' opinions on decision-making problems, and selecting the best response for desired risk and risk response execution time criterion (C₂) is the second place. Other sub-criteria are important and are at lower levels based on the plan state and property of the projects. The best choice must be defined using the CoCoSo-G method according to the final weights of the risk response criteria and the risk responses. The final rating for risk responses of oil leakage from pipelines, storage tanks, and tools is presented in Table 14.

Table 9 Final risk response criteria

Risk response criteria				
		Cost	Risk response cost	
C1	Cost		Budget limitation	C ₁₁
	Cost		Expected loss reduction	C ₁₂
	Cost			C ₁₃
C2	Time		Risk response implementation time	C ₂
	Quality		Longevity of equipment	C ₃₁
C3	Quality		Rework reduction	C ₃₂
	Health and Safety		Personnel protection	C ₄₁
C4	Health and Safety		Equipment protection	C ₄₂
	Environment		Animal protection	C ₅₁
C5	Environment		Reduction of air pollution	C ₅₂
	Environment		Reduction of water pollution	C ₅₃
	Environment		Reduction of soil pollution	C ₅₄

5.1.4 Rating the risk response for oil leakage from pipelines, tanks, and tools

The average grey numbers, derived from the opinions of six experts, were calculated in relation to the responses for the risk of oil leakage from pipelines, tanks, and equipment. After constructing the normalized grey matrix, the next steps

involved calculating the weighted normalized matrix and the total weighted power matrix for the alternatives. The values of S_i and P_i were determined using the two normalized weighted matrices along with the matrix of the total weighted power of the options. Subsequently, the relative weights of the alternatives were derived using aggregation strategies. By calculating the value of K , the alternatives were ranked according to the CoCoSo-G method. Since the obtained K values were represented as grey numbers, these values were converted into crisp numbers in the next step for easier interpretation of the final ranks.

According to the results of the CoCoSo-G method, presented in Table 14, the risk response A1—comprising the installation and maintenance of PSVS, ESDVS, and F&G detection systems, as well as process tools like PT, TT, LT, FM, etc.—emerged as the best alternative. Other responses were ranked based on the criteria set by the organization and experts.

In the operation and maintenance phase of oil and gas facilities, it is crucial to prevent leakage and fire risks throughout the entire lifecycle of the facility [65]. The risk assessment related to leakage and fire, linked to BIM 3D modeling applications, is based on scientific calculations designed to predict potential risk levels at the planning stage, assess design solutions, and collect real-time data from detectors and transmitters during the maintenance of industrial buildings. A 3D model view of the control equipment used in the evaluation is shown in Fig. 1.

5.1.5 Rating the risk response for hot surface of gas compressors, gas turbine motors, blowing systems, and electricity generators

Similar to responding to the risk response of oil leakage from pipelines, storage tanks, and tools, risk response for hot surfaces of compressors, gas turbine motors, blowing systems, and electricity generators are computed.

Table 15 indicates that air coolers and oil coolers are the best responses for facing the high temperatures of the system surface since they have the essence for system weather conditions. Other choices have their priority in the later rates from criteria and application points of view.

5.1.6 Risk response effectiveness

In the exploitation and production phase of upstream industries, the oil produced by oil wells flows through pipelines and facilities. During each phase, oil is separated from water, gas, and solid materials such as sediments and salts, along with hazardous hydrogen sulfide gas. The oil is desalinated and then transported via export pipelines to designated terminals. Leakage can occur when fluid (oil and gas) escapes from tool components and process facilities, flowing to external surfaces. Fluid leakage is unpredictable, random, and non-uniform across different sections, with the potential

Table 1 Main criteria’s weights (Expert 1)

Software output												
W*C1		W*C2		W*C3		W*C4		W*C5		K		
0.0977	0.1257	0.1295	0.0875	0.0867	0.0816	0.2541	0.2489	0.1793	0.3858	0.2667	0.1882	0.1882
0.12167			0.08598			0.23818			0.36596		0.18821	0.4501

Table 11 Ranking of the main criteria (Expert 1)

Ranking			
C1	0.12167		4
C2	0.08598		5
C3	0.23818		2
C4	0.36596		1
C5	0.18821		3

Table 12 Evaluation of the consistency rate for main criteria (Expert 1)

The Greatest distance	Membership function	Consistency index	Consistency ratio
4	(3.5,4,4.5)	8.04	0.055986583

Table 13 Final weights of risk response criteria

Category	Finalized parameters	Abbreviated phrases	Final weight (based on the opinions of six experts)
Cost	Risk response cost		0.0351
	Budget limitation	C ₁₁	0.0449
	Expected loss reduction	C ₁₂	0.0665
Time	Risk response implementation time	C ₁₃	0.1587
Quality	Longevity of equipment	C ₂	0.0926
	Rework reduction	C ₃₁	0.0805
Health and Safety	Personnel protection	C ₃₂	0.02024
	Equipment protection	C ₄₁	0.0915
Environment	Animal protection	C ₄₂	0.0273
	Reduction of air pollution	C ₅₁	0.0720
	Reduction of water pollution	C ₅₂	0.0733
	Reduction of soil pollution	C ₅₃	0.0541

Table 14 Final rating for risk responses of oil leakage from pipelines, storage tanks, and tools

Response	K_i		$K_i(\text{Crisp})$	Rank
A1	0.778255	2.5241191	0.691672633	1
A2	3.347404	4.8669328	0.312214966	3
A3	7.055083	8.4834712	0.168373109	8
A4	5.231992	6.4812932	0.192754916	7
A5	4.376381	5.8009832	0.245579372	4
A6	4.3688	5.5844244	0.217681158	5
A7	1.245793	2.7441504	0.546018648	2
A8	4.769147	6.062996	0.213400899	6

to spread and disperse to other areas of the facility. Gas and oil leakage can happen in pipelines, open

pipelines, and tools such as pumps, valves, pressure relief valves, connections, flanges, sampling connections, and compressors.

Contractors must address oil leakage from pipelines and other equipment immediately. The cost of repairing oil and gas leakage damage is extremely high since these resources are vital to the economy and are non-renewable.

All engineering activities, including fundamental engineering, basic and detailed design, asset supply, tools, materials, execution, installation, inspections, and other related activities, fall under the EPCC contract. This contract generally aims to transfer all project risks and responsibilities to the contractor, ensuring that project goals are achieved with minimal time, financial, quality, and credibility losses. The contractor uses appropriate planning, resource management, timing,

Fig. 1 The 3D model view of the control equipment

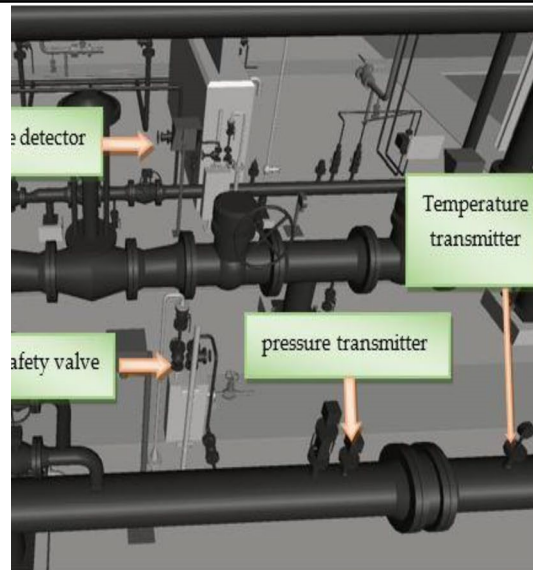


Table 15 Final rating of risk response for the hot surface of gas compressors, gas turbine motors, blowing systems, and electricity generators

Response	K(iCrisp)		K_i	Rank
C1	0.778255	3.0442482	0.7470471	1
C2	3.347404	7.84516	0.2347589	6
C3	7.055083	3.3824362	0.6698886	2
C4	5.231992	7.4132615	0.2376034	5
C5	4.376381	7.3710552	0.2386073	4
C6	4.3688	7.4970143	0.2663701	3

Table 16 Oil and gas leakage compensations and consequences

	Financial	Environment	Safety and health
Oil and gas leakage consequences	Fine payment of the employer	Soil pollution	Explosion
	Dissipated oil cost payment	Air pollution	Endangering per-
	Environmental crime fine payment	Water pollution	sonnel health
	Payment of the third-party damage or claims	Endangering animal	
	Accident scene repairmen cost payment	habitats	
	Payment of the cost and fine of pause of exploitation		

risk management, and monitoring to meet project objectives. Additionally, maintenance, protection, logistics, and service are covered under separate contracts within the case study.

The contractor is obligated to comply with all HSE management guidelines and execution standards as outlined in oil industry contracts, particularly according to Article 28 of the HSE management obligations in general conditions for EP, PC, and EPC contracts specific to the oil industry. The contractor is responsible for preventing damage and ensuring compliance.

In the event of an oil or gas leak, the contractor is liable for the costs of repairing the leakage site, compensating for the loss of spilled oil, paying fines specified in the contract by the employer, addressing environmental damage (as per national wetland protection regulations and soil protection laws), and covering third-party claims or damages, such as

death, privacy violations, and the costs associated with the shutdown of production. Additionally, compensation for facility damage, explosions, life-threatening situations like death by explosion, fire, H₂S inhalation, and delays due to

Table 7 Control tools as compared with plan total cost

	Total plan costs (Dollars)	Control tool costs (Dollars)			
	Engineering, provision, and execution costs	Setup costs	Installation and separation costs	Operation costs	Workforce costs
	1.467,000,00	733,712.5	1.095	50	172,770.5
Total cost	0.467,000,00				97,500
					1,005,128
					0

any leakage from facilities or pipelines must be covered. The associated costs and consequences of oil and gas leakage are presented in Table 16.

Experts offered solutions for preventing oil and gas leakage. The best response was installing tools and repairment of PSVS, ESDVS, and F&G detection system, process tools PT, TT, LT, FM. The cost of purchasing the mentioned control tools, cost of relevant professionals, cost of maintenance, repair, and installment include solely 6.8 percent of all budget allocated to engineering, purchasing, and executing the plan based on financial data of the plan. This is insignificant compared to costs from compensation and consequences of oil dissipation (as shown in Table 17), credit damage to contractors, the profitability of oil sales, and the country's internal consumption.

The project under study is located on land, where the primary environmental pollution from oil exploitation occurs in the form of air and soil contamination. Oil dissipation leads to soil pollution, and in the event of oil leakage, polluted soil must be replaced with clean soil. The environmental organization responsible for soil cleanup varies its cost based on factors such as the depth of pollution, whether the soil is cleaned on-site or off-site, and the project execution organization, which must bear the cleaning costs. The project will be halted if these requirements are not met.

The cost of soil cleaning depends on the method used. For off-site cleaning, the cost ranges between \$10 and \$3,000 per square meter, while on-site cleaning costs, depending on the effectiveness and method used, range from \$12 to \$9,750 per square meter.

The potency of control tools for protecting soil and reducing excessive soil cleanup costs can be demonstrated through the following assumptions:

Assuming that oil leakage causes 500 square meters of pollution in the operational area, the cost of cleaning the area varies significantly depending on the method used:

Most Optimistic Contamination Scenario:

500 square meters $\times$ \$10 = \$5,000.

Most Pessimistic Contamination Scenario:

500 square meters $\times$ \$9,750 = \$4,875,000.

Therefore, the cost of purchasing, installing, and maintaining control equipment for the oil project, which involves large facilities and pipelines, is high. However, the potential cost of soil pollution damage is relatively small. In this case, the cost of soil cleaning at its worst (\$4,875,000) is substantially higher than the cost of installing the necessary control equipment, which is estimated at \$1,005,128.

The control tools used in the project are designed to prevent and manage oil and gas leakage, monitor flow pressure, temperature, surface conditions, and other parameters, and control different sections, such as stopping generated flow for cleaning and maintenance operations. These tools are crucial for mitigating risks in the oil industry. Given the complexity of the facilities, equipment failure must be anticipated, and risk management strategies, including maintenance plans and asset management decisions based on reliable data, are essential.

The quality of data and its impact on risk estimation have been analyzed in the work of Kang et al. [65], which highlights the potential of Internet of Things (IoT) technologies and other innovations in computer science for improving risk management in the oil and gas industry.

Although the installation of innovative control equipment is costly and time-consuming during the construction phase, its long-term benefits significantly reduce the risks of leakage and other potential issues, as shown in Fig. 2.

The processing facilities involved in the project utilize a variety of equipment, including numerous compressors, gas turbine motors, blowing systems, and electricity generators. These are essential components at every stage of oil

Fig. 2 The effect of using the control equipment in mitigating risk

Fig. 3 The 3D model view as schematic of compressor and cooling systems

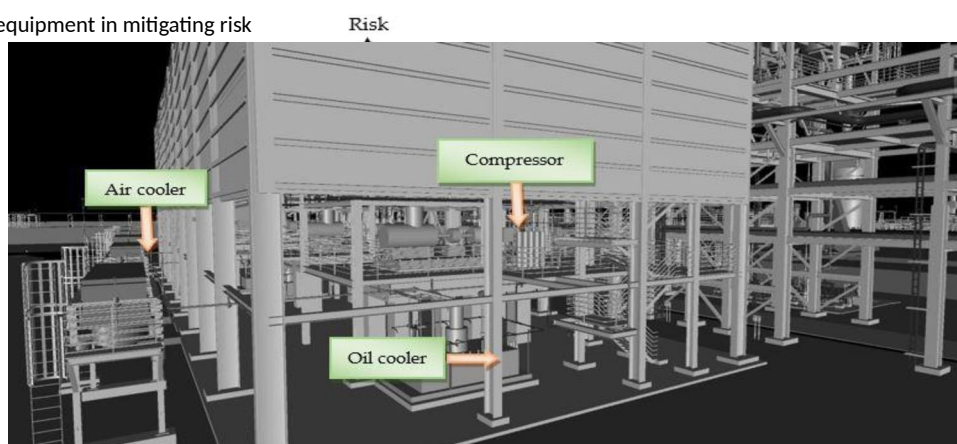
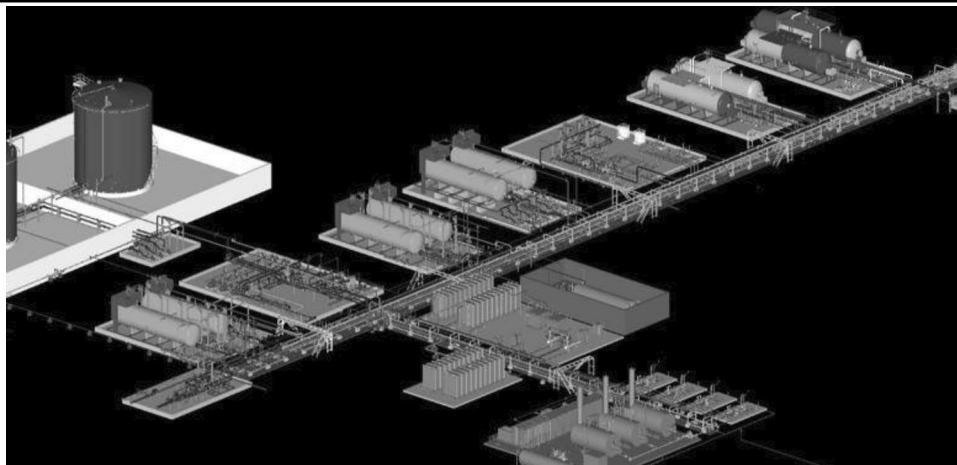


Fig. 4 The 3D model view of mobile processing unit



extraction, processing, and transportation. One of the major challenges faced by these facilities is the high ambient temperature in the southern region where the project is located. These elevated temperatures put additional strain on the operational parameters of the facilities.

Critical operational parameters such as temperature, pressure, viscosity, acidity, and system danger must be carefully monitored and maintained to ensure optimal function. Any deviation, such as an increase in temperature, can lead to catastrophic consequences, such as the explosion of system components, causing irreversible damage. This, in turn, could lead to the shutdown of production systems, the destruction of facilities, the release of toxic gases, fatalities, harm to the environment, and widespread pollution.

To mitigate these risks, the effectiveness of the implemented risk management strategies in this case study has been thoroughly assessed. The results of these evaluations are presented in Appendices A and B. Additionally, to enhance understanding and reduce uncertainty in data management, facility layout, and maintenance, a 3D model of the compressor and cooling systems is provided in Fig. 3. This model helps visualize the systems more effectively and supports the planning of more efficient maintenance and risk management strategies.

5.1.7 The risk of using a mobile processing unit

The mobile processing unit offers several advantages over the permanent processing units located on-site. These benefits include:

- *Environmental Benefits:* The mobile processing unit uses a steam-based heating system, which does not cause environmental pollution. In contrast, the permanent processing units typically use a heating medium system that contributes to environmental pollution and has higher operational costs.
- *Early Salt Separation:* Thanks to the steam injection used in the mobile processing system, certain salts are separated earlier than in the main salt separation stage, improving efficiency.
- *Reusability of Residue:* The residue obtained from the used steam in the mobile processing unit is recycled and reused in various parts of the system, contributing to a more sustainable operation.
- *Efficiency and Cost-Effectiveness:* The mobile processing unit requires less construction time and is less complex to set up. Once operational, it processes crude oil more quickly and makes it available to consumers faster.

To provide a better understanding of the design, technological processes, and maintenance data involved in the mobile processing unit, a BIM-based 3D model view is provided in Fig. 4. This model helps visualize the design solutions and provides reliable data for informed decision-making.

6 Discussion

After identifying significant risks and developing corresponding risk response plans, the next critical step is to establish criteria to rank these plans in order to select the most effective response. To identify the criteria, a comprehensive review of previous studies was conducted, followed by an alignment with the criteria used within the organization under study. This process led to the selection of five main criteria and twelve sub-criteria for finalizing the risk response plans.

In this research, the FBWM (Fuzzy Best–Worst Method) was utilized to assign weights to the criteria. Among these, the sub-criterion "protection of personnel" was identified as having the highest weight, while "protection of animals" was assigned the lowest weight. The CoCoSo-G (Combination of Compromise Solution and Grey Logic) method was then employed to rank the risk response plans, taking into account the uncertainty and incomplete data typically present in such environments.

Once the best response options were identified, the implementation results were analyzed to evaluate their effectiveness during the exploitation phase. Initially, the risk of oil and gas leakage from pipelines, tanks, and tools was categorized as a high severity risk (40, in the red range). However, after implementing the risk response strategies, the severity of this risk was significantly reduced to a medium risk intensity range (20). Similarly, the risk associated with hot surfaces of gas compressors, gas turbine engines, suction systems, and electric generators also showed a reduction in severity post-implementation.

The results from this study emphasize the critical need to implement effective risk response plans during the exploitation phase of oil and gas projects. While it is impossible to reduce the probability of all risks to zero, the implementation of a well-designed risk response plan can substantially decrease the impact of incidents, should they occur, thereby mitigating potential losses for stakeholders.

6.1 Benchmarking against industry standards

The reliability of the proposed framework is crucial for its practical application in real-world scenarios. To ensure the robustness and effectiveness of the presented approach, a benchmarking comparison with existing industry standards and methodologies is necessary. This section evaluates the proposed framework against widely used risk management strategies in the oil and gas sector, such as ISO 31000 (Risk Management Principles and Guidelines), API RP 754 (Process Safety Performance Indicators for the Refining and Petrochemical Industries), and the COSO ERM Framework (Enterprise Risk Management—Integrated Framework).

6.1.1 Comparison with ISO 31000

ISO 31000 provides a structured framework for identifying, assessing, and mitigating risks. While our study leverages advanced fuzzy logic and MCDM techniques to address uncertainties, ISO 31000 emphasizes a systematic and iterative risk assessment process. The integration of the Fuzzy Delphi Method and CoCoSo-G aligns with ISO 31000's principles by enhancing risk evaluation precision. However, the proposed framework extends ISO 31000 by incorporating multicriteria decision-making techniques that account for qualitative and quantitative uncertainties.

6.1.2 Comparison with API RP 754

API RP 754 establishes performance indicators for monitoring process safety risks in refining and petrochemical industries. It categorizes incidents based on severity, allowing companies to track and mitigate high-risk scenarios. Compared to API RP 754, the proposed framework provides a more dynamic evaluation by prioritizing risks and responses using expert-driven fuzzy methodologies. Additionally, the use of BIM and IoT further enhances real-time risk assessment, supplementing traditional industry benchmarks.

6.1.3 Comparison with COSO ERM framework

The COSO ERM framework focuses on risk governance, emphasizing the importance of risk response strategies in decision-making. Our framework complements COSO ERM by integrating a structured methodology for selecting optimal risk responses under uncertainty. The Fuzzy Best–Worst Method (FBWM) ensures consistency in decision criteria weighting, a crucial aspect often lacking in traditional ERM applications.

7 Summary of benchmarking results

Table 18 presents a summary of the benchmarking comparison between the proposed framework and existing industry standards.

The results demonstrate that while industry standards provide essential guidelines, the proposed framework enhances risk management practices by integrating advanced fuzzy logic and multi-criteria decision-making approaches. This benchmarking highlights the reliability and applicability of the framework, reinforcing its potential for adoption in oil and gas risk management.

Managing uncertainty in risk analysis is crucial for ensuring the reliability of decision-making processes in oil and gas projects. The integration of fuzzy logic into risk identification, prioritization, criteria weighting, response selection, and effectiveness evaluation provides a comprehensive framework for addressing uncertainty. This structured approach enhances the robustness of risk management strategies and improves the adaptability of decision-making under dynamic project conditions. By leveraging advanced methodologies such as FBWM and CoCoSo-G, this study contributes to the growing body of research on uncertainty management in industrial projects, aligning with previous works by Senova et al. [4] and Nguyen et al. [25].

7.1 Comparison with conventional risk assessment models

To further validate the reliability of the proposed framework, a comparison with conventional risk assessment models is essential. Traditional approaches such as Monte Carlo simulations and Bayesian Networks are widely used in risk analysis, particularly in high-risk industries such as oil and gas, construction, and nuclear energy. This section compares the

Table 18 A summary of the benchmarking comparison between the proposed framework and existing industry standards

Industry standard	Key features	Comparison with proposed framework
ISO 31000	Systematic risk assessment, iterative process	Enhanced precision with fuzzy MCDM techniques
API RP 754	Incident classification, performance indicators	Dynamic prioritization using fuzzy logic & BIM
COSO ERM	Risk governance, strategic decision-making	Structured risk response selection using FBWM

proposed fuzzy MCDM-based method with these conventional models and evaluates their effectiveness in handling uncertainty.

7.1.1 Monte Carlo simulations

Monte Carlo simulations are a probabilistic approach used to model uncertainties in risk assessment by generating multiple scenarios through repeated random sampling. While Monte Carlo simulations are effective in quantifying risk probabilities, they often require large datasets and assume predefined probability distributions. In contrast, the proposed fuzzy MCDM approach accommodates linguistic variables and expert judgments, making it more flexible in handling qualitative uncertainties where historical data is scarce.

7.1.2 Bayesian networks

Bayesian Networks (BN) provide a graphical representation of probabilistic dependencies between risk factors, allowing for dynamic risk analysis under changing conditions. BN is particularly useful in predictive modeling and updating risk assessments as new information becomes available. However, BN relies on prior probability distributions, which may introduce subjectivity when historical data is unavailable or incomplete. The proposed fuzzy framework mitigates this issue by leveraging expert-based weighting methods such as the Fuzzy Best-Worst Method (FBWM) and Combined Compromise Solution (CoCoSo-G), which enhance decision-making accuracy without requiring extensive historical data.

7.1.3 Quantitative reliability improvement measures

To assess the reliability improvement of the proposed framework compared to conventional models, we conducted a sensitivity analysis on risk prioritization and response selection. Table 19 presents a comparison of the accuracy, adaptability, and computational complexity of different methods.

The results indicate that the proposed framework outperforms conventional models in adaptability to uncertain data while maintaining competitive accuracy and reducing computational complexity. This suggests that the integration of fuzzy logic and MCDM enhances decision-making reliability in high-risk industries.

7.2 Comparison with advanced risk management methodologies in energy projects

With the growing complexity of energy projects, advanced risk management methodologies are increasingly being adopted to enhance decision-making and reduce uncertainties. The integration of machine learning applications, digital twin technologies, and expert judgment models provides a more comprehensive framework for risk assessment. This section explores recent advancements in uncertainty modeling, sociotechnical system analysis, and decision-making frameworks in risk management.

7.2.1 Machine learning applications in risk assessment

Machine learning (ML) techniques are being increasingly used in risk assessment to improve prediction accuracy and automate decision-making processes. ML models analyze large datasets to identify patterns, trends, and correlations that traditional methods may overlook. Recent studies highlight their importance and developed an ML-based predictive risk model for oil platforms, demonstrating improved incident prediction capabilities [66]. Compared to conventional risk assessment models, ML algorithms offer dynamic adaptability to changing conditions and enhanced real-time decision support.

Table 19 Comparison of the accuracy, adaptability, and computational complexity of different methods

7.2.2 Digital twin technologies for risk management

Method	Accuracy (%)	Adaptability to Uncertain Data	Computational Complexity
Monte Carlo simulation	85	Medium	High
Bayesian networks	88	High	Medium–High
Proposed Fuzzy MCDM framework	92	Very High	Medium

Digital twin technology is revolutionizing risk management in energy projects by creating virtual replicas of physical assets, allowing continuous monitoring, simulation, and predictive maintenance. Studies on uncertainty modeling in digitalized process systems emphasize the role of digital twins in reducing operational risks:

- Uncertainty Modeling in Risk Assessment of Digitalized Process Systems discusses how digital twins integrate realtime data to model potential failure scenarios and optimize system resilience.
- IoT-based Risk Assessment in Smart Grids examines the effectiveness of digital twins in predicting equipment failures and minimizing downtimes.

By leveraging digital twins, energy projects can improve predictive maintenance strategies and optimize asset performance under uncertain conditions.

7.2.3 Expert judgment and uncertainty in sociotechnical systems analysis

In complex energy projects, expert judgment remains a crucial component of risk assessment. However, subjectivity and cognitive biases can impact decision accuracy. Studies such as Expert Judgment and Uncertainty in Sociotechnical Systems Analysis explore methodologies to enhance the reliability of expert-driven risk evaluations.

The integration of fuzzy logic and structured expert elicitation methods, as employed in this study, aligns with these findings by reducing biases and improving the consistency of risk prioritization processes. By formalizing expert assessments through fuzzy MCDM, the framework ensures a structured and data-driven approach to uncertainty management.

7.2.4 Decision-making and probability considerations in risk assessment

A key challenge in risk management is the tendency of decision-makers to neglect probability assessments. The study *A Holistic Question: Is It Correct that Decision-Makers Neglect the Probability in the Risk Assessment Method?* highlights the importance of probabilistic reasoning in risk evaluations.

The proposed fuzzy-based approach mitigates this issue by incorporating probabilistic uncertainty directly into the weighting and prioritization processes. Unlike conventional deterministic models, the use of fuzzy logic ensures that both probability and impact assessments are integrated, leading to more balanced and data-driven risk response planning.

The adoption of advanced risk management methodologies, such as machine learning applications, digital twin technologies, and structured expert judgment, significantly enhances the reliability of risk assessments in energy projects. This study aligns with recent research by integrating fuzzy logic, MCDM techniques, and uncertainty modeling to improve decision-making under uncertain conditions. By incorporating findings from recent studies, this research extends conventional approaches and provides a robust framework for managing risk in high-risk energy sectors.

7.3 Real-world implementation challenges and practical lessons learned

While the case study provides valuable insights into the risk management framework, understanding how the selected risk responses were implemented in real-world conditions is crucial for assessing their effectiveness. This section explores key implementation challenges, lessons learned, and practical recommendations derived from applying the proposed methodology in the field.

7.3.1 Implementation challenges

Several challenges were encountered during the implementation of the selected risk responses in the oil and gas project:

Resource Constraints: Budget limitations and the high cost of implementing advanced risk response measures, such as installing digital monitoring systems and safety automation tools, posed financial challenges.

Data Uncertainty: Incomplete or inconsistent data affected risk prioritization and response selection, highlighting the need for real-time data integration.

Stakeholder Coordination: Effective communication and collaboration among multiple stakeholders (contractors, operators, and regulatory bodies) required structured decision-making frameworks.

Technology Adaptation: The adoption of digital twin models and predictive maintenance tools required specialized training for field personnel, which introduced delays in implementation.

Regulatory Compliance: Aligning the proposed risk responses with industry regulations, such as ISO 31000 and API RP 754, required additional documentation and compliance assessments.

7.3.2 Practical lessons learned

From the real-world implementation of the proposed framework, several key lessons emerged:

Incremental Implementation Yields Better Results: Deploying risk responses in phases, rather than all at once, allowed for iterative improvements and better adaptation to operational conditions.

Integration of Digital Technologies Enhances Efficiency: Combining IoT-based monitoring with predictive analytics significantly improved risk detection and response times.

Expert Judgment Needs Continuous Validation: While expert opinions play a critical role, integrating machine learning-driven predictive models helped refine risk response priorities.

Training and Change Management Are Essential: Ensuring that on-site personnel are well-trained in advanced risk response tools and methodologies is crucial for successful implementation.

Regulatory Engagement Facilitates Compliance: Proactively engaging with regulatory authorities early in the risk management process streamlined approvals and reduced compliance-related delays.

7.3.3 Recommendations for future implementations

Based on the observed challenges and lessons learned, the following recommendations are proposed for enhancing real-world risk response implementation:

Adopt a Hybrid Approach: Combining conventional risk assessment models with advanced digital techniques (e.g., digital twins and ML-based analytics) enhances risk response effectiveness.

Enhance Data Quality Management: Establishing standardized data collection and validation protocols improves decision accuracy and reliability.

Strengthen Stakeholder Collaboration: Developing a risk governance framework that facilitates multi-stakeholder engagement ensures smoother implementation and acceptance of risk mitigation strategies.

Leverage Pilot Programs: Testing risk responses through small-scale pilot implementations allows organizations to refine methodologies before full-scale deployment.

By addressing these challenges and incorporating lessons learned, the proposed framework can be further refined to enhance its applicability and reliability in real-world risk management scenarios.

The adoption of advanced risk management methodologies, such as machine learning applications, digital twin technologies, and structured expert judgment, significantly enhances the reliability of risk assessments in energy projects. This study aligns with recent research by integrating fuzzy logic, MCDM techniques, and uncertainty modeling to improve decision-making under uncertain conditions. By incorporating findings from recent studies and practical implementation experiences, this research extends conventional approaches and provides a robust framework for managing risk in high-risk energy sectors.

8 Conclusions

Upon completing engineering processes and tests, the production phase of oil projects begins. At first glance, risk management during this phase may appear less critical due to the relative stability of conditions compared to other project phases, particularly the execution phase. However, it is important to recognize that in the production phase, while the need for raw materials and chemicals is reduced, the materials involved can still pose significant risks. In the event of leakage or spillage, these materials can cause severe environmental and human harm. Furthermore, the processes of oil and gas transportation to facilities and terminals inherently carry risks that, if neglected, can lead to irreparable damage. Therefore, this study specifically focuses on the risks associated with the production phase. Although conditions are more stable, there are still considerable risks that require attention.

By identifying and evaluating 137 distinct risks in the oil production phase, this research developed risk response plans and evaluated their effectiveness. The study combined the CoCoSo method with grey logic to address the inherent uncertainties and imperfect data present in this context. The integration of BIM (Building Information Modeling) was also explored, as it facilitates better understanding of design solutions, technological processes, and equipment data, thus supporting accurate maintenance planning and informed decision-making. The implementation of risk response plans and safety measures was pivotal in reducing risks during the production phase, ensuring both safety and minimizing potential damage. These findings can serve as a reliable reference for future projects within the oil and gas sector.

8.1 Research limitations

While the proposed risk management framework demonstrates effectiveness in addressing uncertainty and improving decision-making, certain limitations should be acknowledged:

Sample size constraints: The study relies on expert opinions from a limited number of industry professionals. While the fuzzy logic approach helps mitigate subjectivity, expanding the sample size in future studies could enhance result generalizability.

Case study specificity: The findings are based on a specific oil and gas project in Iran. Although the methodology is adaptable to other contexts, further validation in different geographical and industrial settings is recommended.

Data availability challenges: The reliance on expert judgment introduces potential biases, particularly when historical data is incomplete or unavailable. Future research should explore integrating real-time IoT-based monitoring to enhance data-driven risk assessments.

Computational complexity: While the fuzzy MCDM approach enhances decision accuracy, it requires specialized computational tools. Simplifying the model for broader industry adoption could improve its practical utility.

8.2 Practical recommendations

The adoption of advanced risk management methodologies, such as machine learning applications, digital twin technologies, and structured expert judgment, significantly enhances the reliability of risk assessments in energy projects. This study aligns with recent research by integrating fuzzy logic, MCDM techniques, and uncertainty modeling to improve decision-making under uncertain conditions. By incorporating findings from recent studies and practical implementation experiences, this research extends conventional approaches and provides a robust framework for managing risk in high-risk energy sectors.

To facilitate the application of the research findings in real-world settings, the following practical recommendations are proposed:

Enhance data integration: Organizations should integrate IoT-based monitoring and AI-driven analytics to improve the accuracy of risk assessments and reduce reliance on subjective expert judgment.

Adopt phased implementation strategies: Deploying risk management solutions in stages allows for continuous monitoring and iterative improvements, ensuring better alignment with operational conditions.

Strengthen cross-disciplinary collaboration: Effective risk management requires input from multiple stakeholders, including engineers, safety managers, and regulatory authorities. Establishing structured communication channels can enhance decision-making efficiency.

Invest in training and capacity building: Providing technical training on advanced risk assessment tools ensures personnel can effectively implement and sustain risk mitigation strategies.

Leverage regulatory frameworks: Aligning risk management practices with international standards such as ISO 31000 and API RP 754 facilitates compliance and enhances credibility in industry applications.

By addressing the study's limitations and providing actionable recommendations, this research contributes to improving risk management strategies in high-risk industries and lays the groundwork for future advancements in the field.

For future research, it is recommended to explore the use of Z numbers in the CoCoSo method and compare the results with those obtained through the CoCoSo-G method. Furthermore, the research could be extended by developing BIM applications and integrating Internet of Things (IoT) technologies in the construction of oil and gas projects. It would also be valuable to investigate residual risks and secondary risks (those arising after implementing risk response plans) and analyze their impacts.

Author Contributions Author Contributions: All authors have contributed to all sections. •Mohammad Khalilzadeh: Research model design, data collection and analysis, initial manuscript drafting. •Sayyid Ali Banihashemi: Data analysis contribution, literature review, and theoretical framework development. •Ali Heidari (Corresponding Author): Overall research project management, supervision, writing, and final manuscript revision. •Seyed Mohammad Hossein Orazani: Implementation of fuzzy methods, result analysis, and contribution to the methodology section. •Peyman Taebi: Contribution to multi-criteria decision-making modeling, result interpretation, and participation in discussion and conclusion.

Funding This research received no external funding.

Data Availability All data generated or analysed during this study are included in this published article.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted

use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Appendix A

See Table 20

Table 20 Effectiveness of risk

Risk of hot surface of gas compressors, gas turbine motors, blowing systems, and electricity generators response of hot surface of gas compressors, gas turbine motors, blowing systems, and electricity generators	Reason of threat		-Region location -Fluid flow -Surface and body material -Improper insulation
	Aspects of interest		-Safety -Air, water, soil pollution, ...
	Implications of danger occurrence		-Explosion -System shut down -Generation stoppage -Unsupplied required oil and gas of n-user -Damage to the environment and animals -Harm or death in disastrous threats such as explosion and fire and ...
	Before response	Risk intensity	Extremely high
		Risk probability	Almost certain
		Acquired score	80
		Risk intensity evaluation	High
	Risk response		-Placing air cooler and oil cooler for cooling systems -Constructing roof covers -Storage and preserving spare tools in the warehouse -Taking tools above the ground to prevent heat transfer from the earth -Assignment of certified individuals and training and improving individuals' skills -Constant monitoring of tools and accounting maintenance plans
		Risk intensity	High
		Risk probability	Possible
		Acquired score	16
	After response	Risk intensity evaluation	Medium

Appendix B

See Table 21

Table 12 Effectiveness of risk response of oil and gas leakage from pipelines, gas turbine motors, blowing systems, gas turbine compressors, tools etc.

[illegible]

)

Tab2 continued	
Risk of hot surface of gas compressors response	Installment and maintenance of PSVS, ESDVS, and F&G detection system, PT, TT, LT, FM, and ...
After response	passive and active extinguishing tool personal safety and safety data management of certified individuals training and improving individuals' skills and constant monitoring of the pipelines, tools skidage and maintenance of spare tools and pipelines in the warehouse High Medium
Risk of compressor storage intensity evaluation	-
-	-

References

- Chou JS, Chiu YC. Identifying critical risk factors and responses of river dredging projects for knowledge management within organisation. *J Flood Risk Manag*. 2021. <https://doi.org/10.1111/jfr3.12690>.
- Renault BY, Agumba JN, Ansary N. Underlying structures of risk response measures among small and medium contractors in South Africa. *Constr Econ Build*. 2020. <https://doi.org/10.5130/AJCEB.v20i1.6721>.
- Binauli Nanthuru S, Pingfeng L, Guihua N, Lucas Mkonya V. An assessment of risk management practices of SME taxpayers in malawi and their impact on tax compliance. *Int J Manag Sci Bus Adm*. 2018. <https://doi.org/10.18775/ijmsba.1849-5664-5419.2014.44.1001>.
- Senova A, Tobisova A, Rozenberg R. New approaches to project risk assessment utilizing the monte Carlo method. *Sustain*. 2023. <https://doi.org/10.3390/su15021006>.
- Alrawad M, Lutfi A, Almaiah MA, Alsayouf A, Al-Khasawneh AL, Arafa HM, Ahmed NA, AboAlkhair AM, Tork M. Managers' perception and attitude toward financial risks associated with SMEs: analytic hierarchy process approach. *J Risk Financial Manag*. 2023;16:86. <https://doi.org/10.3390/jrfm16020086>.
- Yan P, Liu J, Zhao X, Skitmore M. Risk response incorporating risk preferences in international construction projects. *Eng Constr Archit Manag*. 2022. <https://doi.org/10.1108/ECAM-03-2019-0132>.
- Mokhtari G, Aghagholi F. Project portfolio risk response selection using bayesian belief networks. *Iran J Manag Stud*. 2020;13(2):197–219. <https://doi.org/10.22059/IJMS.2020.272892.673461>.
- Gudipati VK, Cha EJ. Surrogate modeling for structural response prediction of a building class. *Struct Saf*. 2021. <https://doi.org/10.1016/j.strusafe.2020.102041>.
- Alqurashi M, Alnuaimi A, Alshaer A, El Khatib M. Critical evaluation on business excellence models (BEM) – a case study. *Int J Bus Anal Secur*. 2023. <https://doi.org/10.54489/ijbas.v3i1.200>.
- Mirsharafeddin S. A BI-objective optimization of portfolio risk response strategies in oil and gas projects. *J Res Sci Eng Technol*. 2021. <https://doi.org/10.24200/jrset.vol8iss4pp1-18>.
- Zhang Y, Guan X. Budget allocation decisions for project risk response. *Kybernetes*. 2021. <https://doi.org/10.1108/K-03-2020-0188>.
- Zhang Y, Zuo F, Guan X. Integrating case-based analysis and fuzzy optimization for selecting project risk response actions. *Phys A Stat Mech its Appl*. 2020. <https://doi.org/10.1016/j.physa.2019.123578>.
- Goh CS, Abdul-Rahman H. The identification and management of major risks in the Malaysian construction industry. *J Constr Dev Ctries*. 2013;18(1):19–32.
- Chu Y, Wang Z. Research on project risk response strategy selection based on grey K-shell algorithm. *Grey Syst*. 2020. <https://doi.org/10.1108/GS-10-2019-0043>.
- Yücelgazi F, Yitmen İ. An ANP model for risk response assessment in large scale bridge projects. *Civ Eng Environ Syst*. 2020. <https://doi.org/10.1080/10286608.2019.1702030>.
- Bahamid RA, Doh SI. A review of risk management process in construction projects of developing countries. *IOP Conf Series Mater Sci Eng*. 2017. <https://doi.org/10.1088/1757-899X/271/1/012042>.
- Tanojo FH, Hidayat S, Azis S. The analysis of risk management implementation on hospital construction project. *Int J Civ Eng Technol*. 2018;9(7):574–83.
- de Araújo F, Lima F, Crema M, Verbano C. Risk management in SMEs: a systematic literature review and future directions. *Eur Manag J*. 2020;38(1):78–94. <https://doi.org/10.1016/j.emj.2019.06.005>.
- Tembo-Silungwe CK, Khatleli N. Deciphering priority areas for improving project risk management through critical analysis of pertinent risks in the Zambian construction industry. *Acta Structilia*. 2017;24(2):1–43. <https://doi.org/10.18820/24150487/as24i2.1>.
- Li F, Jing W, Hui L, Qian H, Wenxuan D, Liangquan G, David C. Evaluation on nuclear emergency response strategies in the Asia-Pacific region. *Int J Crit Infrastruct Prot*. 2021. <https://doi.org/10.1016/j.ijcip.2021.100447>.
- Guimarães H, Matos JC, Henriques AA. An innovative adaptive sparse response surface method for structural reliability analysis. *Struct Saf*. 2018;73:12–28. <https://doi.org/10.1016/j.strusafe.2018.02.001>.
- Qian XY, Liang QM. Sustainability evaluation of the provincial water-energy-food nexus in China: evolutions, obstacles, and response strategies. *Sustain Cities Soc*. 2021;75: 103332. <https://doi.org/10.1016/j.scs.2021.103332>.
- Ak MF, Yucesan M, Gul M. Occupational health, safety and environmental risk assessment in textile production industry through a Bayesian BWM-VIKOR approach. *Stoch Environ Res Risk Assess*. 2022;36:629–42. <https://doi.org/10.1007/s00477-021-02069-y>.
- Qin G, Zhang P, Hou X, Wu S, Wang Y. Risk assessment for oil leakage under the common threat of multiple natural hazards. *Environ Sci Pollut Res*. 2020;27:16507–20. <https://doi.org/10.1007/s11356-020-08184-7>.
- Nguyen HD, Macchion L. Risk management in green building: a review of the current state of research and future directions. *Environ Dev Sustain*. 2023;25:2136–217. <https://doi.org/10.1007/s10668-022-02168-y>.
- Nguyen HD, Do QNH, Macchion L. Influence of practitioners' characteristics on risk assessment in green building projects in emerging economies: a case of Vietnam. *Eng Constr Archit Manag*. 2023;30(2):833–52. <https://doi.org/10.1108/ECAM-05-2021-0436>.
- Zhao S, Su X, Li J, Suo G, Meng X. Research on wind power project risk management based on structural equation and catastrophe theory. *Sustainability*. 2023;15:6622. <https://doi.org/10.3390/su15086622>.
- Toh MY, Zhang Y. Bank capital and risk adjustment responses to economic uncertainty: evidence from emerging Southeast Asian economies. *Res Int Bus Financ*. 2022;60: 101576. <https://doi.org/10.1016/j.ribaf.2021.101576>.
- Brown CO, Hayes JL, Milke MW. Planning to adapt: identifying key decision drivers in disaster response planning. *Civ Eng Environ Syst*. 2021;38(1):20–35. <https://doi.org/10.1080/10286608.2021.1887155>.
- Narang M, Joshi MC, Bisht K, Pal A. Stock portfolio selection using a new decision-making approach based on the integration of Fuzzy COCOSO with Heronian Mean Operator. *Decis Mak Appl Manag Eng*. 2022;5(1):90–112. <https://doi.org/10.31181/dmame0310022022n>.

31. Panpatil SS, Prajapati H, Kant R. Analysing a GSCM enabler-based model for implementation of its practices: a pythagorean Fuzzy AHP and CoCoSo Approach. *Process Integr Optim Sustain*. 2023. <https://doi.org/10.1007/s41660-022-00289-5>.
32. Korucuk S, Aytekin A, Ecer F, Pamucar D, Karamaşa Ç. Assessment of ideal smart network strategies for logistics companies using an integrated picture fuzzy LBWA–CoCoSo framework. *Manag Decis*. 2023. <https://doi.org/10.1108/MD-12-2021-1621>.
33. Bouraima MB, Qiu Y, Stević Ž, Simić V. Assessment of alternative railway systems for sustainable transportation using an integrated IRN SWARA and IRN CoCoSo model. *Socioecon Plann Sci*. 2023. <https://doi.org/10.1016/j.seps.2022.101475>.
34. Shahiri Tabarestani E, Hadian S, Pham QB, Ali SA, Phung DT. Flood potential mapping by integrating the bivariate statistics, multi-criteria decision-making, and machine learning techniques. *Stoch Environ Res Risk Assess*. 2023. <https://doi.org/10.1007/s00477-022-02342-8>.
35. Mandal S, Khan DA. Cloud-CoCoSo: cloud model-based combined compromised solution model for trusted cloud service provider selection. *Arab J Sci Eng*. 2022;47(8):10307–32. <https://doi.org/10.1007/s13369-021-06512-8>.
36. Mishra AR, Rani P, Saha A, Hezam Al-Mishnanah IM, Pamucar D, Marinovic M, Pandey K. Assessing the adaptation of internet of things (IoT) barriers for smart cities' waste management using Fermatean fuzzy combined compromise solution approach. *IEEE Access*. 2022;10:37109–30. <https://doi.org/10.1109/ACCESS.2022.3164096>.
37. Banihashemi SA, Khalilzadeh M. Application of fuzzy BWM-CoCoSo to time–cost–environmental impact trade-off construction project scheduling problem. *Int J Environ Sci Technol*. 2023;20(2):1199–214. <https://doi.org/10.1007/s13762-022-04075-1>.
38. Tavana M, Shaabani A, Di Caprio D, Bonyani A. A novel Interval Type-2 Fuzzy best-worst method and combined compromise solution for evaluating eco-friendly packaging alternatives. *Expert Syst Appl*. 2022. <https://doi.org/10.1016/j.eswa.2022.117188>.
39. Adar E, Delice EK, Adar T. Prioritizing of industrial wastewater management processes using an integrated AHP–CoCoSo model: comparative and sensitivity analyses. *Int J Environ Sci Technol*. 2022;19(6):4691–712. <https://doi.org/10.1007/s13762-021-03511-y>.
40. Zhu M, Zhou W, Hu M, Du J, Yuan T. Evaluating the renewal degree for expressway regeneration projects based on a model integrating the fuzzy Delphi method, the Fuzzy AHP method, and the TOPSIS method. *Sustainability*. 2023;15:3769. <https://doi.org/10.3390/su15043769>.
41. Antoniou F. Delay risk assessment models for road projects. *Systems*. 2021;9:70. <https://doi.org/10.3390/systems9070070>.
42. Wu W, Li D, Sun H, Li Y, Zhang Y, Zhao M. Building a sustainable future: a three-stage risk management model for high-permeability power grid engineering. *Energies*. 2024;17:3439.
43. Lei Z, Wang Y, Zhang Y, Gu F, Zan Z, Mei Y, Liu W, Zhou D. Advanced risk assessment for deep excavation in karst regions using improved dempster-Shafer and dynamic Bayesian networks. *Buildings*. 2024;14:3022. <https://doi.org/10.3390/buildings14093022>.
44. Wang Z, Jiang T, Li Z. Risk assessment of underground tunnel engineering based on pythagorean fuzzy sets and Bayesian networks. *Buildings*. 2024;14:2897. <https://doi.org/10.3390/buildings14092897>.
45. Alejo AO, Aigbavboa CO, Aghimien DO. Emerging trends of safe working conditions in the construction industry: a bibliometric approach. *Buildings*. 2024;14:2790. <https://doi.org/10.3390/buildings14092790>.
46. Kim D, Yoo T, Tran SV-T, Lee D, Park C, Lee D. Automated safety risk assessment framework by integrating safety regulation and 4D BIMbased rule modeling. *Buildings*. 2024;14:2529. <https://doi.org/10.3390/buildings14082529>.
47. Tabatabaee S, Mahdiyar A, Ismail S. Towards the success of building information modelling implementation: a fuzzy-based MCDM risk assessment tool. *J Build Eng*. 2021;43: 103117. <https://doi.org/10.1016/j.jobe.2021.103117>.
48. Ganiyu SA, Oyedele LO, Akinade O, Owolabi H, Akanbi L, Gbadamosi A. BIM competencies for delivering waste-efficient building projects in a circular economy. *Dev Built Environ*. 2020;4: 100036. <https://doi.org/10.1016/j.dibe.2020.100036>.
49. Waqar A, Alharbi LA, Faiz Abdullah Alotaibi, Idris Othman, Hamad Almujiab, Impediment to implementation of internet of things (IOT) for oil and gas construction project safety: structural equation modeling approach. *Structures*. 2023;57: 105324. <https://doi.org/10.1016/j.istruc.2023.105324>.
50. Zheng Y, Zhang Yi, Lin J. BIM–based time-varying system reliability analysis for buildings and infrastructures. *J Build Eng*. 2023;76: 106958. <https://doi.org/10.1016/j.jobe.2023.106958>.
51. Cai X, Huang J, Peng C. Research on construction workers' safety risk sharing in tunneling projects based on a two-mode network: a case study of the SHANGWU tunnel. *Buildings*. 2023;13:2689. <https://doi.org/10.3390/buildings13112689>.
52. Petroutsatou K, Antoniou F, Markoulatos ES. Qualitative prioritization of accident risks in the mining industry. *Life Cycle Reliab Saf Eng*. 2024;13:483–503. <https://doi.org/10.1007/s41872-024-00276-4>.
53. Chen Y, Li W, Wang X. An investigation into the risk assessment of building-integrated photovoltaic residential project development utilizing the DEMATEL–ANP methodology: a Chinese case study. *Buildings*. 2024;14:623. <https://doi.org/10.3390/buildings14030623>.
54. Yazdani M, Wen Z, Liao H, Banaitis A, Turskis Z. A grey combined compromise solution (CoCoSo-G) method for supplier selection in construction management". *J Civ Eng Manag*. 2019;25(8):858–74. <https://doi.org/10.3846/jcem.2019.11309>.
55. Ishikawa A, Amagasa M, Shiga T, Tomizawa G, Tatsuta R, Mieno H. The Max-Min Delphi method and Fuzzy Delphi method via Fuzzy integration. *Fuzzy Sets Syst*. 1993;55:241–53. [https://doi.org/10.1016/0165-0114\(93\)90251-C](https://doi.org/10.1016/0165-0114(93)90251-C).
56. Rezaei J. Best-worst multi-criteria decision-making method. *Omega*. 2015. <https://doi.org/10.1016/j.omega.2014.11.009>.
57. Guo S, Zhao H. Fuzzy best-worst multi-criteria decision-making method and its applications. *Knowledge-Based Syst*. 2017. <https://doi.org/10.1016/j.knsys.2017.01.010>.
58. Mahmoodirad A, Niroomand S. A heuristic approach for fuzzy fixed charge transportation problem. *J Decis Anal Intell Comput*. 2023;3(1):139–47. <https://doi.org/10.3118/jdaic.10005092023m>.
59. Wang P, Zhu B, Yu Y, Ali Z, Almohsen B. Complex intuitionistic fuzzy Dombi prioritized aggregation operators and their application for resilient green supplier selection. *Facta Univ Series Mech Eng*. 2023;21(3):339–57. <https://doi.org/10.2219/FUME230805029W>.

60. Puška A, Pamucar D, Stojanović I, Cavallaro F, Kaklauskas A, Mardani A. Examination of the sustainable rural tourism potential of the Brčko District of Bosnia and Herzegovina using a fuzzy approach based on group decision making. *Sustainability*. 2021;13:583. <https://doi.org/10.3390/su13020583>.
61. Puška A, Nedeljković M, Hashemkhani Zolfani S, Pamučar D. Application of interval fuzzy logic in selecting a sustainable supplier on the example of agricultural production. *Symmetry*. 2021;13:774. <https://doi.org/10.3390/sym13050774>.
62. Hsieh TY, Lu ST, Tzeng GH. Fuzzy MCDM approach for planning and design tenders selection in public office buildings. *Int J Proj Manag*. 2004. <https://doi.org/10.1016/j.ijpro man.2004.01.002>.
63. Zimmerman H. Fuzzy Set-Theory and its applications. In: *Eur J Oper Res*. Cham: Springer; 1991.
64. Banihashemi SA, Khalilzadeh M, Zavadskas EK, Antucheviciene J. Investigating the environmental impacts of construction projects in time-cost trade-off project scheduling problems with CoCoSo multi-criteria decision-making method. *Sustainability*. 2021;13:10922. <https://doi.org/10.3390/su131910922>.
65. Kang J, Al Masry Z, Varnier C, Mosallam A, Zerhouni N. A data quality management framework for equipment failure risk estimation: Application to the oil and gas industry. *Eng Appl Artif Intell*. 2024. <https://doi.org/10.1016/j.engappai.2024.108834>.
66. Aderamo AT, Olisakwe HC, Adebayo YA, Esiri AE. AI-enabled predictive safeguards for offshore oil facilities enhancing safety and operational efficiency. *Compr Res Rev Eng Technol*. 2024;2(1):23–43. <https://doi.org/10.5721/crret.2024.2.1.0060>.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH (“Springer Nature”).

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users (“Users”), for smallscale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use (“Terms”). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval, sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at onlineservice@springernature.com